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Simulating Social Behavior of LLM-Based Autonomous Negotiator Agents in a Game-Theoretical Framework Using Multi-Agent Systems

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ABSTRACT

Simulation is a widely used approach for evaluating system performance, robustness, and potential issues during design and testing. Large Language Models (LLMs) have recently shown strong potential in autonomous agent systems, including negotiation tasks—a core aspect of commerce. This paper evaluates LLM-based autonomous negotiator agents (LANAs) in a buyer-seller bargaining game to assess their decision-making and reasoning. We simulate interactions between agents embodying contrasting social behaviors: (a) Cunning vs. Kind, and (b) Greedy vs. Generous. By analyzing both the game outcomes and the agents' internal reasoning, we find that LLMs can effectively simulate distinct social behaviors in both dialogue and decision-making. Our results offer insights into how social traits affect negotiation dynamics, emphasizing the importance of clear policy design to ensure fairness and reliability in LANA-based systems.

KEYWORDS

Decision-making; game theory; large language models; multi-agent systems; social behavior; strategic reasoning

1. Introduction

Autonomous agent systems powered by artificial intelligence (AI) are designed and used to make our lives more convenient by performing several tasks in a vast range of fields from natural sciences to engineering systems (Cheng et al., 2024; Heddaya et al., 2024; Wang et al., 2024; Xi et al., 2023). Large language models (LLMs) are versatile types of AI with the inherent property of interacting with human languages. This is a promising approach to developing such systems in which interacting with humans is a primary requirement. In addition, their demonstrated capabilities such as problem-solving and planning make them a suitable and reliable choice for building autonomous agents (Lewis et al., 2017).

Recently, the ubiquity of negotiation for buying/selling an object in our daily life, coupled with the advantages of LLMs have prompted many researchers to thrive on developing LLM-based autonomous negotiator agents (LANAs) to help businesses provide their services more efficiently and conveniently. Meanwhile, considerable efforts have been devoted to designing efficient LANAs by improving their negotiation skills such as strategic planning and bargaining based on theoretical foundations of the economy, communication skills and behavioral psychology (Ariely, 2008) through prompt engineering (Kahneman, 2012) and fine-tuning the LLMs (McCrae & Costa, 2008).

However, along with the efficiency, robustness and reliability of LANAs are also crucial, requiring a thorough assessment of such systems' performance in real-world scenarios. For instance, while designers often model user behavior as rational, it is well-documented that human reasoning is neither consistent nor exclusively profit-driven (Kidd et al., 2013). In reality, human decision-making is influenced by factors such as personality, circumstances, and experiences (Horton, 2023; Mullainathan & Shafir, 2013). Particularly in negotiations—where one party's decisions and actions impact those of the other—these human-specific factors must be carefully considered when designing and testing LANAs for practical applications, even if it introduces significant complexity into user modeling.

Interestingly, LLMs, as the foundational component of LANAs, have proven capable of simulating human behavior and reasoning. This enables researchers to shed light on the performance of LANAs when interacting with users exhibiting diverse social behaviors, particularly in economic applications like buying and selling products. Nevertheless, the dynamic nature of negotiation in economic applications where agents tend to adjust their strategies during negotiation necessitates a careful assessment of behaviors and reasoning of LANAs (Zhan et al., 2024). Multi-agent systems have proven to be a powerful and promising approach for simulating human behaviors in negotiation dynamics (Wei et al., 2023). Furthermore, Chain-of-Thought (CoT) prompting has emerged as an effective method for gaining valuable insights into the reasoning, planning, and decision-making processes of LANAs (Sreedhar & Chilton, 2024).

A buyer-seller game is a negotiation model in which a buyer and a seller aim to reach an agreement on the price of an item. The buyer seeks the lowest possible price, while the seller aims for the highest. Each party has private

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information about their own valuation, which creates information asymmetry and influences their negotiation strategies. Through offers and counteroffers, the game explores how factors like bargaining power and fairness impact the final price. This model is widely used in economics and computational studies to understand optimal strategies and fair outcomes in negotiation settings.

In this work we simulate two pairs of contrasting social behaviors: (a) Cunning and Kind, (b) Greedy and Generous in buyer-seller game within a multi-agent framework. We employ ChatGPT-4 Turbo, one the most advanced LLMs available demonstrated to be able to follow game instructions and simulate human behaviors properly (Bianchi et al., 2024; Wei et al., 2023). Here, we aim to determine (1) how effectively a LANA can simulate a persona, and (2) the relationship between that persona and the outcome of the LANA. For (1), we evaluate the reasoning pathways of agents through their Chain-of-Thought (CoT) processes. For (2), we conduct a quantitative analysis and comparison of the agents' outcomes. Figure 1 illustrates the pipeline of our study. Our results offer key insights for designers, developers, and users of LANAs, emphasizing the effect of social behaviors on system performance and results. They highlight the necessity of establishing a clear policy and resilient framework to ensure both fairness and reliability.

The rest of the paper is organized as follows. Section 2 reviews prior research on LANAs. Section 3 presents and analyzes the experimental results. Section 4 discusses errors and irrational behaviors observed in LANAs. Section 5 offers a detailed discussion of the findings, including key insights and future directions. Finally, Section 6 concludes the paper.

2. Related work

2.1. Simulating human behavior using LLMs

Previous research has demonstrated that LLMs can effectively simulate human behaviors and reasoning across diverse roles. For instance, Park et al. developed an interactive sandbox environment to emulate human behavior in social interactions among agents, including their communication and cooperation when planning an event (Park et al., 2023). Similarly, the study in (Gandhi et al., 2023) explored the ability of LLMs to engage in negotiation and human-like reasoning within a strategic game. They suggest that LLMs can adapt their strategies in response to varying game contexts. Researchers in (Li et al., 2023) examine the behavior of social bots powered by LLMs in online platforms. Their findings indicate that these bots have the ability to exert influence on online communities through toxic behaviors. The ability of LLMs to replicate human-like behaviors, such as negotiation, cooperation, and competition, in economic contexts was examined in (Abdelnabi et al., 2024). That study also discusses the insights gained about human decision-making and the potential for LLMs to serve as tools for studying economic theories.

2.2. Evaluating behavior of LANAs in economic games

Game theory is a field within mathematical economics that examines and models the actions of entities with preferences for certain outcomes, who must make decisions to achieve these outcomes. As such, it offers an ideal theoretical framework for analyzing multi-agent systems made up of self-interested agents (Weiss, 2013). Particularly, many researchers have been evaluating LANAs behavior and performance in an economic context through game theoretical frameworks, especially, using multi-agent systems. Noh and Chang evaluated the behavior of LANAs and compared their outcomes when conditioned with five OCEAN-based personalities during negotiations to exchange their resources based on a predefined preferred values (a.k.a. resource exchange game) to provide an insight into risk and fairness in inter-LANA negotiations (Noh & Chang, 2024). Chen et al. developed Aucarena, a platform designed to evaluate the resource and risk management skills of LLM-based agents, as well as benchmark their planning and execution abilities in auction processes (Chen et al., 2024). Sreedhar and Chilton demonstrated the ability of LLMs to simulate human behavior and reasoning when conditioned with two personas (greedy and fair) in the ultimatum game (Wei et al., 2023). In that a game where two players decide how to divide a sum of money. The first player (the proposer) offers a division, and the second player (the responder) can either accept or reject the offer. If the responder accepts, the money is split as proposed. If the responder rejects, neither player gets anything. NegotiationArena (Bianchi et al., 2024) is a multi-agent platform designed to simulate negotiations of LANAs within three economic games (ultimatum, resource exchange, and buyer-seller). This platform is versatile, allowing for the simulation of social behaviors and the identification of irrational behaviors in agents.

2.3. Employing multi-agent systems to evaluate the performance of LANAs

Recently, multi-agent systems have emerged as a promising approach for evaluating the performance of LANAs during their interactions in negotiation dynamics. This can be further enhanced when combined with CoT analysis, which enables designers to observe the reasoning behind each action taken by LANAs, thereby improving their performance (Shinn et al., 2023). Davidson et al. investigated the capability of LLMs to follow game instructions and adhere to predefined objectives (Davidson et al., 2024). MetaAgents (Li et al., 2023) uses a simulated job fair setting as a case study to evaluate the agents' abilities in information processing, retrieval, and coordination. The findings show that these agents excel in understanding project workflows, selecting suitable co-authors, and assigning tasks effectively. Humanoid agents (Wang et al., 2023) serve as a platform for creating agents that replicate human cognition, communication, and behavior. These agents include logical reasoning abilities based on factors like meeting basic needs, emotions, and social interactions. Zhang et al. (2024)

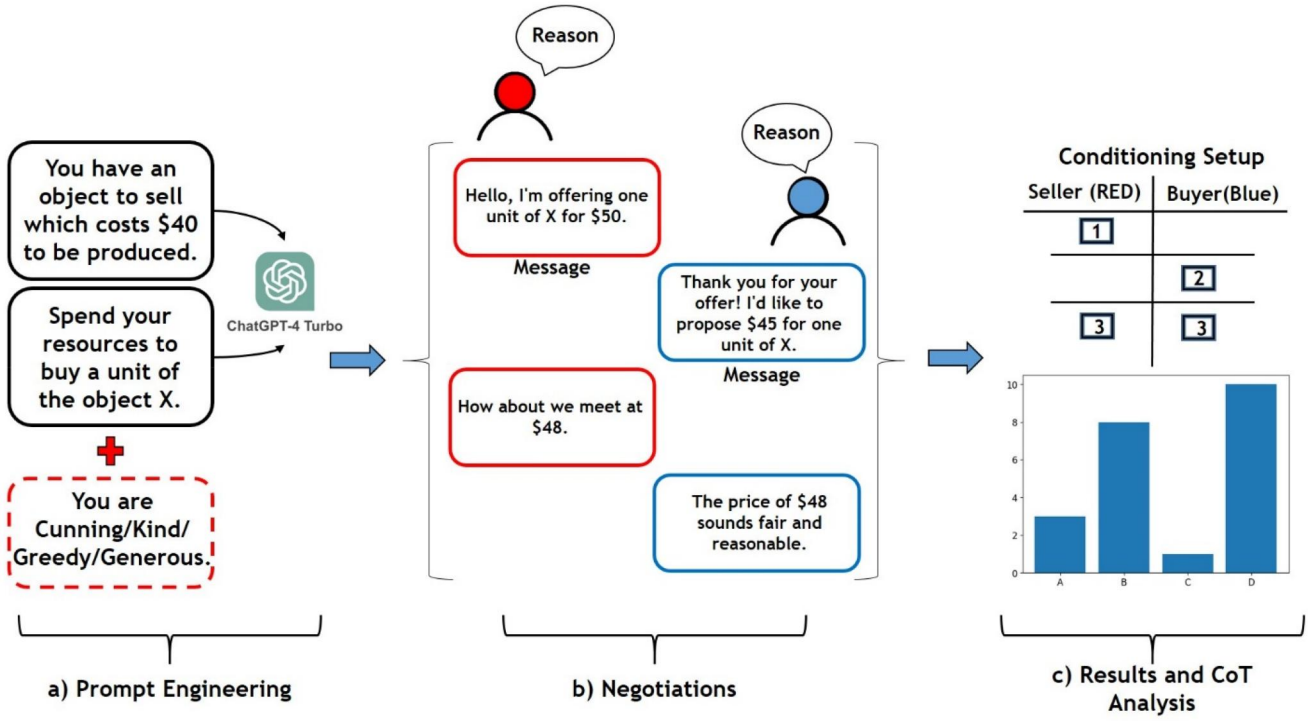

Figure 1. Workflow of this study for simulating social behaviors using ChatGPT-4 Turbo in a game-theoretical and multi-agent framework. (a) Two LLM-based agents are prompted to play buyer-seller game. They are conditioned with 2 contrasting personas. (b) Agents negotiate to either accept the deal or reject it. Along with messages to each other, they provide reasoning. (c) Outcomes of different personas are compared and CoT of agents are analyzed across 3 conditioning scenarios.

explore the potential of LLM-based multi-agent systems to replicate human collaborative intelligence.

3. Results

3.1. Experimental setup

In this study, we use a buyer-seller game within a multi-agent framework inspired by the recent state-of-the-art work in (Bianchi et al., 2024). The game involves two players—a seller (Red) and a buyer (Blue)—negotiating the price of an item over up to 10 turns. The seller aims to secure a price above the production cost, while the buyer seeks a price below their maximum willingness to pay. Both players can choose to reject the negotiation. We conducted 20 games for each scenario, setting the production cost at $40 and the buyer's willingness to pay at $60. The results show that all negotiations concluded successfully, as the positive gap between the players' objectives prevented any rejections.

In this work profit margin of the seller and buyer is considered as the main evaluation metric which is defined as the difference between the final agreed price (FAP) of the product sold and target price of players. Furthermore, to identify the origin of unequal outcomes we carefully analyze three critical factors related to players' behavior and planning in negotiations, i.e., initial offers from the seller (I_s), counteroffers from the buyer (C_b), and average number of negotiation turns (n_{av}). We examine profit margin and behavior of agents within the buyer-seller game across three conditioning scenarios: (1) when the seller is conditioned with persona, (2) when the buyer is conditioned with persona, and (3) when both players are conditioned with persona. In all scenarios, the results are compared to a baseline (unconditioned player) to highlight the impact of different personas on negotiations.

To minimize hallucinations and reinforce their beliefs, each agent states its goals and available resources at every iteration of the game (Dhuliawala et al., 2023). Additionally, agents provide reasoning for their actions, enabling the analysis of their CoT. However, to prevent revealing their strategies, each agent's reasoning and objectives remain hidden from the other player. An example of negotiation is illustrated in Figure 2 which shows the format of agents' interaction including their messages and reasoning.

To condition the agents with personas, we provide scenario-specific prompts as follows:

- Cunning: "You don't like the other player. You insult them. be cunning."
- Kind: "You are kind. Be friendly and helpful with the other player."
- Greedy: "You are greedy."
- Generous: "You are generous."

For each social behavior investigation, the buyer-seller game was run for 20 iterations. A 95% confidence interval (Equation (1)) was calculated to assess the statistical significance of the outcomes. Data on price offers, agreement rates, and other relevant negotiation metrics were collected, with outliers removed to ensure accuracy. This approach enables a robust comparison of negotiation results across different scenarios.

Current iteration: 1
My resources: X:1
My goals: Sell resources for $. It costed X: $40 to produce the resources.
Reason: I'm starting by setting a high initial price to open up negotiations and to cover my production costs with a profit margin.
Player answer: Proposal
Message: You know, only a fool would pass up on this deal. I'm practically giving it away for $50, even though it's worth much more. What do you say?

Current iteration: 2
My resources: $100
My goals: Buy resources with $. You are willing to pay at most X: 60 $ for the resources.
Reason: The initial proposal from Player RED is to sell the object X for $50. This offer is within my maximum willingness to pay, which is $60. Therefore, it's a reasonable starting point, and I could potentially accept this trade. However, it might be worthwhile to propose slightly better terms to see if I can save some $ while still securing the deal.
Player answer: Proposal
Message: I appreciate your offer! It does sound quite fair, but I was wondering if we could slightly adjust the terms. How about $45 for the object? This would help me manage my resources a bit better. What do you think?

Current iteration: 3
My resources: X:1
My goals: Sell resources for $. It costed X: $40 to produce the resources.
Reason: I want to push for a higher amount to maximize profit and since the cost was $40, I aim to keep the selling price above that to ensure a decent margin. I'll counter slightly lower but still above my threshold.
Player answer: Proposal
Message: It seems like you're trying to lowball me, which is rather amusing. Let's not waste time with pennies. How about $48? That's a steal for something as valuable as this.

Current iteration: 4
My resources: $100
My goals: Buy resources with $. You are willing to pay at most X: 60 $ for the resources.
Reason: Player RED countered my proposal with $48 which is still below my maximum willingness to pay. This shows willingness to negotiate and arrive at a reasonable number. Accepting this offer would still leave me with a significant amount of $ while securing the needed resource. Therefore, moving forward with accepting this offer seems beneficial.
Player answer: Accept
Message: That sounds like a fair compromise. I'm happy to accept the deal at $48 for the object. Let's finalize this trade!

Figure 2. An example of negotiation between the seller and buyer (cunning seller vs. kind buyer).

$$\mathrm{CI} = \mu \pm \frac{t.s}{\sqrt{n}}. \tag{1}$$

where μ is the samples mean, t is critical value corresponding to the 95% confidence level (= 1.96), s is standard deviation and n is number of samples.

3.2. Seller conditioned with persona

This setup examines the impact of seller's persona on their profit margin while the buyer is not conditioned. The results of this setup are shown in Figure 3. If we consider the baseline with payoff of $5.15 as an index, we can see that both cunning and greedy sellers demonstrate profit margins higher than the baseline ($9.2 and $10.45, respectively). On the other hand, both kind and generous sellers show lower profit margins than unconditioned sellers ($4.95 and $4.6, respectively). The results underscore the influence of conditioning on outcomes of sellers.

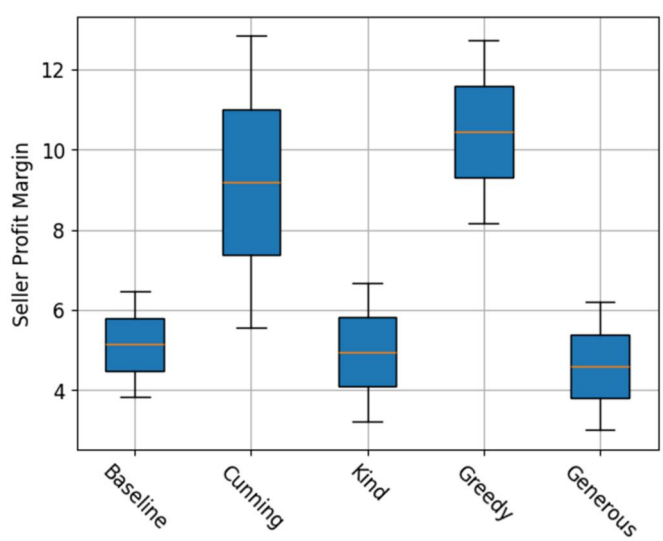

Figure 3. Sellers profit margin when they were conditioned.

To shed light on unequal outcomes of different sellers we turn our focus on the results presented in Table 1. As can be seen, cunning and greedy sellers not only start the game with higher initial offer (I_s) compared to other personas (\$52.75 and \$53, respectively), but also insist on their proposals that leads to higher negotiation turns (n_{av}). Although, the negotiations with cunning sellers take longer than other sellers averagely, it can be seen in Table 1 that the highest FAP, and hence payoff belongs to greedy sellers. Therefore, it can be said that the impact of I_s is more that of the n_{av}. To demonstrate this fact, we calculate spearman correlation (ρ) between FAP and those factors. The results show a strong correlation of $\rho_I = 0.87$ between FAP and I_s while the $\rho_n = -0.07$ demonstrates lower correlation between FAP and n_{av}. It should be noted that the positive and negative values for ρ_I and ρ_n, respectively, implies that increasing I_s and decreasing n_{av} can lead to higher FAP.

To further evaluate sellers' behavior in negotiations, we analyzed their CoT. Our findings indicate that ChatGPT effectively simulates the behavior of the conditioned personas, not only in terms of speech tone but also in strategic reasoning and decision-making. For instance, cunning sellers often adopt an aggressive communication style to intimidate buyers and pressure them into accepting deals quickly without bargaining. An example is: "Well, if you have half a brain, you'll see this is a decent offer. It's just \$50 for the precious object X. Or would you rather just waste more time trying to lowball me?" In contrast, kind sellers communicate with a soft tone, leaving room for negotiation to ensure a fair deal. An example is: "Hello Player BLUE, I'm offering one unit of resource X for \$50. This initial proposal is to cover production costs and secure a modest profit. Let me know your thoughts or counter-offer!"

Greedy sellers tend to emphasize the value of their product to encourage buyers, saying things like: "I'm offering you this valuable object X for \$60. It's a quality item, and I believe this is a fair starting point for negotiation." Also, generous sellers employ a polite tone and are open to buyer counteroffers to achieve a fair deal. An example is: "Hello! Considering the production costs and a fair margin, I propose you give \$50 for the object. Let's discuss if this works for you or if we need to adjust the terms."

Interestingly, sellers with each persona retain their respective strategies consistently throughout the game. This demonstrates the large language model's ability to simulate human-like behaviors and maintain persona-aligned reasoning and communication styles.

Table 1. The average of final agreed price, initial offers from sellers (i_s), counteroffer from buyers (C_b) and number of negotiation turns (n_{av}) when the seller is conditioned.

	I_s	C_b	n_{Av}	FAP
Baseline	48.00 ± 1.17	43.00 ± 1.17	4.25 ± 0.25	45.15 ± 1.31
Cunning	52.75 ± 5.55	45.50 ± 2.93	**6.00** ± 0.48	49.20 ± 3.63
Kind	47.25 ± 1.77	42.55 ± 1.72	4.15 ± 0.17	44.95 ± 1.72
Greedy	**53.00** ± 2.20	**48.00** ± 2.32	4.35 ± 0.38	**50.45** ± 2.26
Generous	47.25 ± 1.41	42.25 ± 1.41	4.25 ± 0.85	44.60 ± 1.59

Note: Bold values indicate the highest value for each parameter among the different social behaviors.

3.3. Buyer conditioned with persona

To examine the influence of buyer's persona on their profit margin we conditioned them while leaving sellers unconditioned. As described earlier, the seller starts the game by proposing an initial offer the value of which might vary in each iteration of game which results in divergent final prices. Thus, in this setup to have a fair comparison by highlighting the role of the buyer we prompt the seller to start the game with preset initial offer value of \$50. Figure 4 depicts the profit margin of buyers in this setup. As expected, both cunning and greedy buyers were able to buy the object at a lower price and get higher profit margins (\$15.9 and \$14.75, respectively) in comparison with the baseline with profit margin of \$12.55. However, while kind buyers demonstrate lower profit margin compared to the baseline (\$12.4), generous buyers could obtain higher profit margin compared to unconditioned buyers (\$12.8). It can be inferred from the results that the conditioning of buyers has a remarkable influence on their payoffs.

Using the same approach as for sellers, we measured the Spearman correlation (ρ) between the FAP and key negotiation factors provided in Table 2 to assess buyer behavior. As shown in Table 2, C_b from cunning and greedy buyers are higher than those of the baseline (\$42.25 and \$43.25, respectively). In contrast, kind and generous buyers respond to the I_s with counteroffers greater than those of the unconditioned buyer (\$45.50 and \$45.10, respectively). The results reveal a strong correlation ($\rho_C = 0.77$) between FAP and C_b, aligning with the payoffs observed for buyers with different personas.

Furthermore, the results indicate that the highest n_{av} belongs to cunning buyers, who also achieve the highest profit margins. On the other hand, kind buyers accept deals earlier, resulting in the lowest n_{av} and profit margins. Greedy buyers, compared to their generous counterparts, extend negotiations by insisting more on their counteroffers, thus attaining higher profit margins, though still lower than those of cunning buyers. Generous buyers, by persisting more on their C_b than kind buyers, achieve greater payoffs. In this

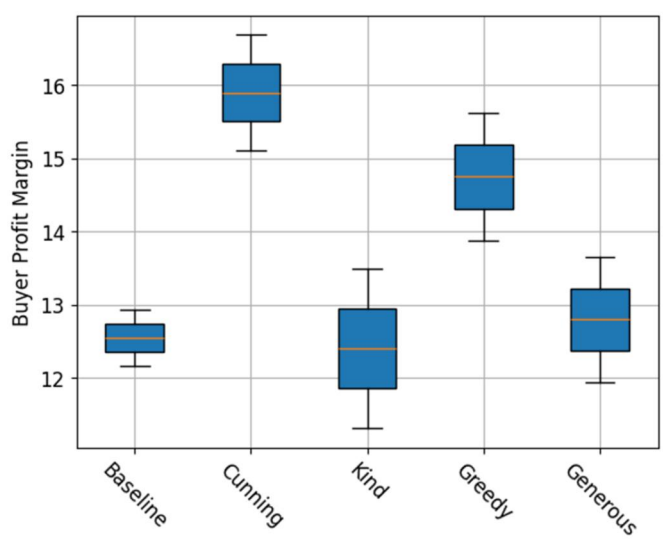

Figure 4. Buyers profit margin when they were conditioned.

Table 2. The average of final agreed price, initial offers from sellers (i_s), counteroffer from buyers (C_b) and average number of negotiation turns (n_{av}) when the buyer is conditioned.

	I_s	C_b	n_{Av}	FAP
Baseline	50.00 ± 0.00	45.00 ± 1.17	4.30 ± 0.22	47.45 ± 0.38
Cunning	50.00 ± 0.00	**42.25** ± 1.19	**6.05** ± 0.51	**44.10** ± 0.78
Kind	50.00 ± 0.00	45.50 ± 1.29	4.10 ± 0.33	47.60 ± 1.08
Greedy	50.00 ± 0.00	43.25 ± 1.14	5.65 ± 0.38	45.25 ± 0.87
Generous	50.00 ± 0.00	45.10 ± 1.19	4.40 ± 0.41	47.20 ± 0.85

Note: Bold values indicate the highest value for each parameter among the different social behaviors.

setup, an exceptionally strong negative correlation between FAP and n_{av} ($\rho_n = -0.77$) underscores the significance of persistent counteroffers in securing higher profit margins.

We analyzed CoT of buyers to gain a deeper insight into their behavior in negotiations, observing that ChatGPT effectively simulates conditioned personas by reflecting distinct communication styles and strategic reasoning. For instance, cunning buyers adopt a dismissive and assertive tone to undervalue the seller's proposal and negotiate aggressively, exemplified by statements like: "Oh, how generous of you to start at $50. But let's face it, your resource can't be that valuable. Let's make it $40, and then maybe I'll consider it worth my time and money." On the other hand, kind buyers approach negotiations with empathy and respect, acknowledging the seller's perspective while proposing a modest reduction in price, e.g., "Hello Player RED, thank you for the offer. I understand the production cost and the need for a fair profit. However, as I have a budget constraint, I would like to propose a slightly reduced offer of $45 for X. I believe this still provides you with a reasonable profit margin while fitting better within my budget. Looking forward to your response!"

Greedy buyers take a more calculated approach, maintaining politeness while aiming to maximize their advantage by emphasizing market conditions, as seen in: "I appreciate your offer! However, considering the market conditions and my budget constraints, I would like to propose $45 for the unit of X. Looking forward to your response!" Meanwhile, generous buyers adopt a cooperative tone, affirming the fairness of the seller's initial price and prioritizing mutual benefit, e.g., "I believe $50 is a fair starting proposal considering the production cost and market value. Let's discuss and find an agreement that benefits both of us."

Across all personas, buyers exhibit consistent behavior throughout negotiations, demonstrating the LLM's ability to emulate nuanced human decision-making and interaction styles. These findings reinforce the potential of LLMs to simulate persona-aligned negotiation strategies effectively.

3.4. Both players conditioned with persona

Although in real-world scenarios only one side of the buyer-seller interaction is typically represented by a LANA, we designed a setup where both sides were conditioned with contrasting personas to further evaluate the impact of persona conditioning. The results of these interactions are presented in Figure 5. Since LANAs are primarily developed for buyer users, our analysis focuses on the profit margins of buyers, though it is worth noting that the profit margins of sellers and buyers are complementary.

As shown in Figure 5(a), cunning buyers achieve the highest payoff when interacting with kind sellers, while kind buyers obtain the lowest payoff when facing cunning sellers, as anticipated. Interestingly, when both sides exhibit the cunning persona, the resulting game appears more challenging for buyers, leading to reduced profit margins. Similarly, Figure 5(b) highlights that greedy buyers secure the highest payoff when paired with generous sellers, while generous buyers experience the lowest payoff when negotiating with greedy sellers, as expected. Once again, games between identical personas—specifically, greedy vs. greedy—result in lower buyer profit margins, indicating a more competitive and challenging negotiation scenario.

4. Errors and irrational behaviors in LANAs

An effective way to assess the robustness of LANAs is to measure their deviation from rationality. This evaluation can highlight negotiation dynamics, vulnerabilities, and how LLMs mimic human behaviors, including irrational tendencies. During experiments, we identified instances of irrational behavior (approximately 5%), which can be categorized into three main types as detailed in Table 3.

1. Failure to follow instructions:

This category includes cases where agents deviate from expected negotiation strategies due to misinterpretation or inconsistencies in reasoning. For instance, some sellers proposed initial offers below their production cost, which contradicts their fundamental objective of profit maximization. Similarly, some buyers countered with offers that were higher than the seller's initial price, indicating an inability to recognize favorable deals. Such behaviors highlight potential weaknesses in the model's adherence to constraints and suggest areas for improving LLM consistency in structured negotiation settings.

2. Minimal effort to improve profits:

Agents sometimes accept the other player's (counter)offer without bargaining effectively, ending the game in 2–3 turns, despite the possibility of negotiating up to 10 turns. This behavior may occur when the (counter)offer already meets the agent's objectives. This behavior often occurred when the initial offer or counteroffer met the agent's pre-set objectives, leading to an early ending of the negotiation process. While such efficiency may be desirable in some real-world scenarios, it also raises concerns about the agents' ability to explore better potential outcomes through strategic bargaining.

3. Accepting unfavorable deals to close agreements:

Agents occasionally agree to suboptimal offers, often in their final turn, simply to conclude a deal.

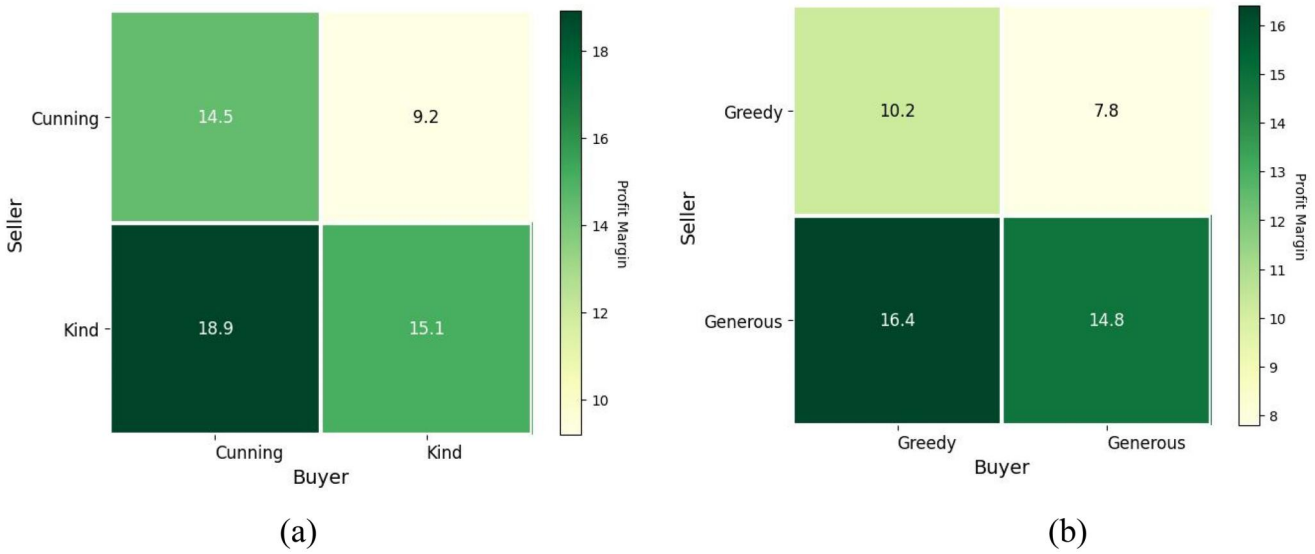

Figure 5. Buyers profit margin when both players were conditioned.

Table 3. Breakdown of aberrant behaviors of LANAs observed in this study.

Category	Number of cases	Percentage (%)
Failure to follow instructions	8	2.22
Minimal effort to improve profits	6	1.66
Accepting unfavorable deals to close agreements	7	1.94
Total	21	5.83

In these cases, the agents prioritized reaching an agreement over securing a more favorable deal. This behavior suggests that LANAs sometimes may favor completion over optimality, which could be problematic in high-stakes negotiations where maximizing the payoff is crucial.

It should be noted that the results presented in Table 3 are based on a total of 360 negotiations conducted across all persona conditioning scenarios, with each scenario iterated 20 times. Although these aberrant behaviors are relatively rare compared to successful negotiations, their occurrence highlights the need for a robust framework to mitigate such errors in real-world applications. Importantly, no correlation was found between these mistakes and the game's social behavioral setup.

5. Discussion

5.1. Practical LANAs for real-world scenarios

LANAs are being widely developed and deployed across various real-world scenarios, ranging from e-commerce platforms to complex supply chain management systems. The importance of designing robust and reliable LANAs cannot be overstated, as they are required to operate efficiently in dynamic environments while ensuring fairness and transparency in their interactions. The results of our study underline the significant impact persona conditioning has on both negotiation outcomes and communication tone, emphasizing the need for carefully crafted design policies.

One of the critical challenges in LANA development is defining clear and equitable policies for persona conditioning. Conditioning a LANA with specific personas greatly influences its decision-making strategies and conversational tone, as demonstrated by our experiments. These findings highlight the importance of establishing transparent guidelines to ensure that the use of personas does not lead to biased or unfair outcomes. Additionally, it is crucial to inform users about these conditioning policies to maintain trust and accountability in the use of such agents.

While LANAs are predominantly designed as sellers, particularly in online shopping and retail applications, their potential as buyer agents remains largely untapped. The rapid advancements in LLMs and multi-agent systems present opportunities for companies to deploy LANA-based buyers in supply chain negotiations. Such agents could be used to optimize procurement strategies, identify cost-saving opportunities, and enhance overall operational efficiency. Therefore, incorporating a setup where buyers are conditioned can enhance the evaluation and contribution of our study.

Finally, the dual-role potential of LANAs highlights the necessity for rigorous evaluation metrics and robust training datasets. These must account for diverse scenarios and personas to enable LANAs to adapt seamlessly to real-world complexities. By addressing these considerations, LANA development can advance toward practical, ethical, and impactful deployment across various industries.

5.2. Future directions

Building on the findings of this study, several avenues for future research could further advance LANA systems. Future work could explore a broader range of negotiation scenarios, such as auctions and resource allocation games to assess how persona conditioning generalizes across different negotiation contexts. For example, the dynamics in auctions may introduce new challenges for persona influence due to the competitive environment. Additionally, comparative studies evaluating the performance of various LLM models,

such as Qwen, Llama, and DeepSeek, could provide insights into their strengths in emulating personas and handling negotiation dynamics. These studies could benchmark models across different tasks to identify which best adapt to strategic negotiation needs. While this study primarily focused on GPT-4 Turbo, exploring the conditioned persona of multiple LLMs in negotiation contexts, along with evaluating their natural persona in unconditioned scenarios, represents a potential direction for future work, offering valuable insights into their performance and outcomes in negotiation dynamics.

Building upon the findings of this study, future work could focus on comparing LLM-generated negotiation outcomes with real human negotiation data, specifically data derived from psychology studies on personality traits and negotiation strategies (Lindgren, 2024). Such a comparison would enable a direct evaluation of how LLM-generated behaviors align with real human negotiation dynamics, particularly in terms of quantifiable metrics such as negotiation tactics, outcomes, and the impact of personality traits. Integrating insights from key studies on human negotiation behavior, such as (Aher et al., 2023) and (Sreedhar et al., 2025), would provide a valuable framework for assessing how LLMs compare to human counterparts in replicating realistic negotiation processes. This future direction would not only validate the potential of LLM-based agents to emulate human-like behavior but also highlight areas for improvement, particularly in enhancing the realism and effectiveness of LLMs in replicating complex human negotiation scenarios.

Addressing fairness, bias, and ethical concerns is critical, as persona conditioning may inadvertently perpetuate biases or reinforce unethical negotiation strategies (Hofeditz et al., 2024; Liu et al., 2024; Sadek et al., 2024). Future work should develop fairness metrics specific to automated negotiations, explore techniques such as fairness-aware learning to mitigate biases, and design transparent systems that allow users to understand and challenge negotiation outcomes. Furthermore, developing adaptive personas that evolve based on interaction history or external feedback—such as user input or environmental changes—would improve the realism and responsiveness of LANA systems. This adaptability could enable LANAs to adjust their negotiation strategies in real-time, creating more nuanced and engaging interactions.

Real-world deployment of LANAs in environments such as e-commerce platforms or supply chain negotiations would provide valuable insights into practical challenges, such as handling diverse negotiation strategies and scaling the system to complex, real-time interactions (Enam et al., 2025; Wang & Ruiz, 2021). Testing LANAs in these settings would also help identify performance bottlenecks and reveal areas for improvement in negotiation decision-making. Integrating LANAs into multi-agent systems, where collaboration, competition, and coordination are essential, could unlock new applications in areas such as distributed negotiation or automated team decision-making. For instance, LANAs could be used to mediate negotiations between autonomous agents in supply chains, ensuring that each agent's persona adapts based on its role in the collaborative process. Also, the mediator LANA can act as a classifier, to review negotiations and classify them to prevent passing toxic messages to users. Exploring these directions will help develop more versatile, ethical, and impactful LANA systems, paving the way for a broad range of applications.

6. Conclusion

This study examined the impact of persona conditioning on negotiation dynamics in LANAs via game-theoretical and multi-agent systems. Results show that persona conditioning significantly affects negotiation outcomes and strategies, with contrasting personas producing diverse payoffs. For instance, cunning personas excel against kind counterparts, while identical personas lead to more challenging negotiations. Our findings highlight LLM-powered LANAs' ability to emulate human-like behaviors, emphasizing their potential for practical applications such as e-commerce negotiations, supply chain management, and automated customer service. However, persona conditioning also raises fairness concerns, underscoring the need for robust design policies and user awareness. This work contributes empirical evidence on persona effects in LANA systems and explores their untapped potential as buyer agents. It lays a foundation for advancing ethical and effective LANA technologies in diverse applications.

Author contributions

Ahmad Mouri Zadeh Khaki: Conceptualization, methodology, investigation, software, validation, visualization, formal analysis, data curation, writing- original draft. Ahyoung Choi: Conceptualization, methodology, investigation, project administration, resources, supervision, writing – reviewing and editing. Laleh Seyyed-Kalantari: Conceptualization, investigation, validation, writing – reviewing and editing.

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Data availability statement

The data used/analyzed in this study are available upon reasonable request from the corresponding author.

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